



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
GB/T 16483-2008, GB/T 17519-2013

**Product Name** Silver Reagent Concentrate

**Revision date** 23-Dec-2020

**Revision Number** 2

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

**Product Name** Silver Reagent Concentrate

**Catalogue Number(s)** 1610445, 1610445EDU

### Other means of identification

**Pure substance/mixture** Mixture

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science  
Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Ltd.  
1st and 2nd Floor, Lumpini 1 Building  
239/2, Rajdamri Road, Lumpini,  
Pathumwan, Bangkok 10330  
Thailand

**Technical Service** +66 2 652 8313  
ctsthailand@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Hong Kong: 800-968-793

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

## SECTION 2: Hazards identification

### **Emergency Overview**

Irritating to skin  
Irritating to eyes

**Appearance** clear liquid

**Physical state** Liquid

**Odour** Odourless

### Classification of the substance or mixture

|                                                              |             |
|--------------------------------------------------------------|-------------|
| <b>Skin corrosion/irritation</b>                             | Category 2  |
| <b>Serious eye damage/eye irritation</b>                     | Category 2A |
| <b>Hazardous to the Aquatic Environment - Acute Hazard</b>   | Category 3  |
| <b>Hazardous to the Aquatic Environment - Chronic Hazard</b> | Category 3  |

Revision date 23-Dec-2020

Review Date

23-Dec-2020

### Label elements



Signal word

Warning

#### **Hazard statements**

Causes skin irritation

Causes serious eye irritation

Harmful to aquatic life with long lasting effects

#### **Precautionary statements**

##### **Prevention**

Wash face, hands and any exposed skin thoroughly after handling

Avoid release to the environment

Wear protective gloves/protective clothing/eye protection/face protection

IF ON SKIN: Wash with plenty of water and soap

If skin irritation occurs: Get medical advice/attention

Take off all contaminated clothing and wash it before reuse

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

If eye irritation persists: Get medical advice/attention

##### **Disposal**

Dispose of contents/container to an approved waste disposal plant

#### **Physical and chemical hazards**

Not applicable.

#### **Health hazards**

Immediate Health Effects: Causes skin irritation (pain, redness and swelling). Causes severe irritation (tears, blurred vision and redness). Irritating, but will not permanently injure eye tissue.

Chronic effects: Not applicable.

#### **Environmental hazards**

This material is a water pollutant. Keep out of drains, sewers, ditches and waterways. Minimise use of water to prevent environmental contamination

#### **Other hazards which do not result in classification**

Not applicable

## **SECTION 3: Composition/information on ingredients**

### **Substance**

Not applicable.

### **Mixture**

| Chemical name  | Weight-% | CAS No    |
|----------------|----------|-----------|
| Silver nitrate | 1 - 2.5  | 7761-88-8 |

Revision date

23-Dec-2020

Review Date

23-Dec-2020

## SECTION 4: First aid measures

### Description of necessary first aid measures

|                                                                           |                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>                                                     | Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                                                            |
| <b>Inhalation</b>                                                         | Get medical attention immediately if symptoms occur. Remove to fresh air.                                                                                                                                                                                                           |
| <b>Eye contact</b>                                                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Keep eye wide open while rinsing. Do not rub affected area. Get medical attention if irritation develops and persists. |
| <b>Skin contact</b>                                                       | Wash off immediately with soap and plenty of water for at least 15 minutes. Get medical attention if irritation develops and persists.                                                                                                                                              |
| <b>Ingestion</b>                                                          | Clean mouth with water and drink afterwards plenty of water. Never give anything by mouth to an unconscious person. Do NOT induce vomiting. Call a doctor.                                                                                                                          |
| <b><u>Most important symptoms and effects, both acute and delayed</u></b> | May cause redness and tearing of the eyes. Burning sensation.                                                                                                                                                                                                                       |
| <b><u>For emergency responders</u></b>                                    | Avoid contact with skin, eyes or clothing. Wear personal protective clothing (see section 8).                                                                                                                                                                                       |
| <b><u>Note to doctors</u></b>                                             | Treat symptomatically.                                                                                                                                                                                                                                                              |

## SECTION 5: Firefighting measures

### Extinguishing media

|                                                            |                                                                                                                                    |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>                        | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| <b>Unsuitable extinguishing media</b>                      | No information available.                                                                                                          |
| <b><u>Specific hazards arising from the chemical</u></b>   | No information available.                                                                                                          |
| <b><u>Special protective actions for fire-fighters</u></b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

## SECTION 6: Accidental release measures

### Personal precautions, protective equipment and emergency procedures

|                                                                    |                                                                                                                        |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions</b>                                        | Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Use personal protective equipment as required. |
| <b>Other information</b>                                           | Refer to protective measures listed in Sections 7 and 8.                                                               |
| <b>For emergency responders</b>                                    | Use personal protection recommended in Section 8.                                                                      |
| <b><u>Environmental precautions</u></b>                            | Prevent further leakage or spillage if safe to do so.                                                                  |
| <b><u>Methods and material for containment and cleaning up</u></b> | Prevent further leakage or spillage if safe to do so. Pick up and transfer to properly labelled containers.            |
| Revision date                                                      | 23-Dec-2020                                                                                                            |

**Precautions to prevent secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## SECTION 7: Handling and storage

**Precautions for safe handling** Take off contaminated clothing and wash it before reuse. Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product. Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product. See Section 8 for information on appropriate personal protective equipment.

**Conditions for safe storage, including any incompatibilities** Keep containers tightly closed in a dry, cool and well-ventilated place. Store according to product and label instructions.

**Incompatible materials** Strong acids. Strong bases. Strong oxidising agents.

## SECTION 8: Exposure controls/personal protection

### Occupational exposure limits

| Chemical name              | China | ACGIH TLV                      |
|----------------------------|-------|--------------------------------|
| Silver nitrate - 7761-88-8 | -     | TWA: 0.01 mg/m <sup>3</sup> Ag |

**Note** See section 16 for terms and abbreviations

### Biological occupational exposure limits

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Monitoring and observation processes

No applicable information was found.

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

**Eye/face protection** If splashes are likely to occur, wear safety glasses with side-shields.

**Skin and body protection** Long sleeved clothing. Wear suitable protective clothing.

**Hand protection** Impervious gloves. Wear suitable gloves.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product.

## SECTION 9: Physical and chemical properties

Revision date 23-Dec-2020

**Information on basic physical and chemical properties**

|                 |                          |
|-----------------|--------------------------|
| Appearance      | clear liquid             |
| Colour          | colourless               |
| Physical state  | Liquid                   |
| Odour           | Odourless                |
| Odour threshold | No information available |

| Property                               | Values                   | Remarks • Method |
|----------------------------------------|--------------------------|------------------|
| pH                                     | No information available | None known       |
| Melting point / freezing point         | 0 °C                     |                  |
| Boiling point / boiling range          | No data available        | None known       |
| Flash point                            | No data available        | Not applicable   |
| Evaporation rate                       | No data available        | None known       |
| Flammability (solid, gas)              | No data available        | None known       |
| Flammability Limit in Air              |                          | None known       |
| Upper flammability or explosive limits | No data available        |                  |
| Lower flammability or explosive limits | No data available        |                  |
| Vapour pressure                        | No data available        | None known       |
| Vapour density                         | No data available        | None known       |
| Relative density                       | No data available        | None known       |
| Water solubility                       | Miscible in water        |                  |
| Solubility(ies)                        | No data available        | None known       |
| Partition coefficient                  | No data available        | None known       |
| Autoignition temperature               | No data available        | None known       |
| Decomposition temperature              | No data available        | None known       |
| Kinematic viscosity                    | No data available        | None known       |
| Dynamic viscosity                      | No data available        | None known       |

**Additional information**

|                      |                |
|----------------------|----------------|
| Explosive properties | Not applicable |
| Oxidising properties | Not applicable |
| Liquid Density       | 1.013          |

**SECTION 10: Stability and reactivity**

**Stability** Stable under normal conditions.

**Possibility of hazardous reactions** None under normal processing.

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Conditions to avoid** None known based on information supplied.

**Incompatible materials** Strong acids. Strong bases. Strong oxidising agents.

**Hazardous decomposition products** None known based on information supplied.

**SECTION 11: Toxicological information**

**Acute toxicity**

**Numerical measures of toxicity**

The following values are calculated based on chapter 3.1 of the GHS document

Revision date 23-Dec-2020

ATEmix (oral) 71,090.90 mg/kg

#### Component Information

| Chemical name  | Oral LD50            | Dermal LD50 | Inhalation LC50 |
|----------------|----------------------|-------------|-----------------|
| Silver nitrate | = 1173 mg/kg ( Rat ) | -           | -               |

**Skin corrosion/irritation** Classification based on data available for ingredients. Irritating to skin.

**Serious eye damage/eye irritation** Classification based on data available for ingredients. Causes serious eye irritation.

**Respiratory or skin sensitisation** Based on available data, the classification criteria are not met.

**Germ cell mutagenicity** Based on available data, the classification criteria are not met.

**Carcinogenicity** .

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Chemical name  | China | IARC     |
|----------------|-------|----------|
| Silver nitrate | -     | Group 2A |

#### Legend

**IARC (International Agency for Research on Cancer)**

Group 2A - Probably Carcinogenic to Humans

**Reproductive toxicity** Based on available data, the classification criteria are not met.

**Specific target organ toxicity — single exposure** Based on available data, the classification criteria are not met.

**Specific target organ toxicity — repeated exposure** Based on available data, the classification criteria are not met.

**Target organ effects** Eyes. Skin. Nasal Septum.

**Aspiration hazard** Based on available data, the classification criteria are not met.

## SECTION 12: Ecological information

**Ecotoxicity** Harmful to aquatic life with long lasting effects.

**Unknown aquatic toxicity** 0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment

| Chemical name  | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                | Crustacea                                                                                                                                 |
|----------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Silver nitrate | -                    | LC50: 0.001339 - 0.001637mg/L (96h, Oncorhynchus mykiss)<br>LC50: 0.00181 - 0.00214mg/L (96h, Pimephales promelas)<br>LC50: 0.00452 - 0.00638mg/L (96h, Pimephales promelas)<br>LC50: 0.00512 - 0.00787mg/L (96h, Poecilia reticulata)<br>LC50: 0.0064 - 0.0106mg/L | EC50: 0.0008 - 0.001mg/L (48h, Daphnia magna)<br>EC50: 0.0008 - 0.0011mg/L (48h, Daphnia magna)<br>EC50: =0.0006mg/L (48h, Daphnia magna) |

Revision date 23-Dec-2020

|  |  |                                                                                                                                                                                                                                                                                                                                                                                                                      |  |
|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  |  | (96h, Pimephales promelas)<br>LC50: 0.00839 - 0.1802mg/L<br>(96h, Oncorhynchus mykiss)<br>LC50: 0.009 - 0.02mg/L (96h,<br>Lepomis macrochirus)<br>LC50: 0.0242 - 0.0484mg/L<br>(96h, Lepomis macrochirus)<br>LC50: 0.05 - 0.07mg/L (96h,<br>Lepomis macrochirus)<br>LC50: =0.0027mg/L (96h,<br>Cyprinus carpio)<br>LC50: =0.0075mg/L (96h,<br>Oncorhynchus mykiss)<br>LC50: =0.009mg/L (96h,<br>Pimephales promelas) |  |
|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

**Persistence and degradability** No information available.

**Bioaccumulative potential** No information available.

**Mobility in soil** No information available.

### SECTION 13: Disposal considerations

**Waste chemicals** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

### SECTION 14: Transport information

**IMDG** Not regulated  
**Transport in bulk according to Annex II of MARPOL and the IBC Code** No information available

**IATA** Not regulated

**China** Not regulated

**Special precautions for user**  
Please refer to the applicable dangerous goods regulations for additional information

### SECTION 15: Regulatory information

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

#### **National regulations**

**Law of the People's Republic of China on Prevention and Control of Occupational Diseases**

Catalogue of occupational hazard factors: Not applicable.  
Catalogue of occupational diseases: Not applicable.

#### **Regulations on the Control over Safety of Hazardous Chemicals**

Revision date 23-Dec-2020

**Product Name** Silver Reagent Concentrate  
**Revision date** 23-Dec-2020

**(M)SDS Number**

HRLS00038

Inventory of hazardous chemicals

The table below indicates ingredients above the cut-off threshold considered as relevant which are listed.  
Weight-% 2

| Chemical name  | Inventory of hazardous chemicals |
|----------------|----------------------------------|
| Silver nitrate | Listed                           |

GB 18218-2009 Identification of major hazard installations for dangerous chemicals

Not applicable

**List of hazardous chemicals under priority management**

Not applicable

**Regulations on Labour Protection in Workplaces Where Toxic Substances Are Used**

Inventory of highly toxic goods

Not applicable

**Regulations for Environmental Management on the First Import of Chemicals and the Import and Export of Toxic Chemicals**

List of toxic chemicals severely restricted for import and export in China

Not applicable

**Measures for the Environmental Management of New Chemical Substances**

**IECSC - China Inventory of Existing Chemical Substances** Contact supplier for inventory compliance status.

International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**SECTION 16: Other information**

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 23-Dec-2020

**Revision Note** \*\*\* Indicates this information has changed since the previous revision.

**Abbreviations and acronyms**

Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AELG(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)

Revision date 23-Dec-2020



**Product Name** Silver Reagent Concentrate  
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HRLS00038

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National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Disclaimer**

**The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text**

**End of Safety Data Sheet**

Revision date 23-Dec-2020

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Review Date

23-Dec-2020